

Class 30 Limits and Continuity

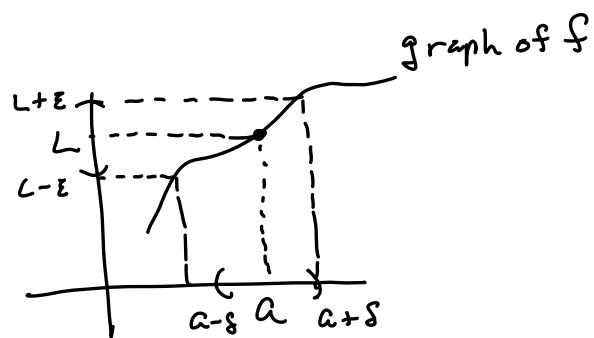
Goals

1. Be able to state Cauchy's definition of limit.
2. Be able to use Cauchy's definition to verify the value of a given limit.
3. Prove a function is continuous at a given point using the definition.

The concept of a limit is crucial to any rigorous treatment of calculus. French mathematician Augustin-Louis Cauchy formed the first rigorous definition in the early 1800's. We still use this definition today. It is not an easy definition to use but with patience and perseverance you will master it. It has never been easy. When Cauchy taught it to his students, they rioted!

Cauchy's Definition of Limit

Here is a picture of the situation.



Example 1. Use Cauchy's definition to prove $\lim_{x \rightarrow a} (mx + b) = ma + b$

where $m \neq 0$

Example 2. Use Cauchy's definition to prove $\lim_{x \rightarrow 5} \frac{1}{x} = \frac{1}{5}$

Intuitive idea of limit: $\lim_{x \rightarrow a} f(x)$ is the number that $f(x)$ gets close to as x gets close to a .

See example of finding limits from the graph on page 267 and exercises 1-6 on page 279.

Continuity

Definition: A function, f , is continuous at $x=a$ if

Theorem: If f is continuous at $x=a$ and k is any constant, then $g(x) = k f(x)$ is continuous at $x=a$.

If $k = 0$, $g(x) = k f(x)$ is just the constant function $g(x) = 0$.

For homework, prove any constant function $f(x) = c$ is continuous at every $x = a$.